

HYPER RAG diagnosis model with concept detection, identifying underlying biological mechanisms

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MPhil Project Proposal

Abstract

Traditional diagnostic approaches often focus on treating symptoms without identifying underlying causes. This project aims to develop a verifiable method for diagnosing root causes by identifying relationships and causations between diseases and symptoms that are frequently overlooked. We propose leveraging the explainability of concepts, as demonstrated in [5], to validate diagnoses and integrating hypergraphs within a BERT model [1] to capture higher-order relationships across medical literature. This approach seeks to uncover previously unseen connections, such as the recently identified link between celiac disease and schizophrenia [2]. These insights highlight the importance of interdisciplinary correlations in advancing diagnostic accuracy. In the medical field people often have one to two specialties causing medical professionals to become pigeonholed in their diagnostic abilities. This project aims to address this.

The project will emphasize Retrieval-Augmented Generation (RAG), global concept interpretability, and the integration of Hypergraph Neural Networks (HGNNs) [1] within our RAG system to enhance the discovery of higher-order relationships. The integration of hypergraphs into the RAG system allows concept detection to be carried out as node analysis is necessary for concept detection. Ultimately, this approach aims to support more comprehensive, causative diagnostics for challenging symptoms across medical fields.

1 Introduction, Approach, and Outcomes

In recent years, machine learning has become an essential tool in medical diagnostics, with applications like machine vision commonly used to identify physiological abnormalities. However, current diagnostic models lack reliable verification mechanisms and may overlook complex patterns linking symptoms to underlying diseases, contributing to misdiagnoses. To put this in context the

scale of this issue misdiagnosis is a huge cost to every health system worldwide with major diagnostics errors found in 10% to 20% percent if autopsies [4].

This research addresses these challenges by proposing a Retrieval-Augmented Generation (RAG) diagnostic model using HyperBERT [1] as the retrieval engine. HyperBERT employs blocks that are a combination of semantic information combined with an alignment function. This system allows us to represent hierarchical relationships within medical concepts.

A critical feature of this model is its verification mechanism through concept detection as seen in [5].

By analyzing and grouping node activations within the HyperBERT layers which are a combination of both semantic and structural information we hope to identify the highest scoring concepts. These are a set of neurons which globally explain the behavior for retrieving documents. These identified concepts will be used later for validation once the LLM has generated a diagnosis.

The retrieved information is then passed to a large language model (LLM) such as GPT-4, or Med-PaLM-2 to generate a diagnosis. The goal is to identify the causative biological mechanisms leading to symptoms. To verify this diagnosis, the model cross-references the detected concepts during retrieval, ensuring consistency. It also ensures the practitioner can double check said sources to ensure the suggested treatment won't result in unintended consequences.

The outcome is a robust, interpretable, and verifiable diagnostic tool capable of accurately identifying both common and rare health conditions, thus supporting more reliable and comprehensive medical diagnostics.

The goal of this thesis is to:

- Organise a training dataset, and a test dataset where the diagnosis is known
- Start prototyping a simple RAG system with a BERT model.
- Build HyperBert into the retrieval part of the RAG system
- Implement concept detection into hyperBert blocks.
- Compare diagnosis generated with concepts detected.
- Fine tune both the HyperBert model and the LLM to be as accurate and understandable as possible.
- Test if the model identifies previously unknown /known concepts/underlying biological mechanisms.
- Deliver a novel model for diagnosis that escapes the traditional issues with the use of LLM in medical Diagnosis.

2 Workplan

- **Finalize the literature review (Timeline: December 7th - December 21st)**

Assess the state-of-the-art literature and thoroughly understand how concept detection will be applied to the HyperBERT blocks. This includes an assessment of state-of-the-art concept detection as seen in [5] and methodologies from [3].

- **Prototyping model (Timeline: December 22nd – January 4th)**
Set up a RAG (Retrieval-Augmented Generation) system for generating diagnoses, and implement a retrieval system using HyperBERT with concept detection applied within the HGNN integrated into the semantic blocks.
- **Tuning of the HyperBERT model (Timeline: January 5th – February 1st)**
Once the initial prototype is complete, start tuning HyperBERT to improve its effectiveness in recognizing complex, higher-order patterns in challenging symptoms.
- **First Results (Timeline: February 2nd – February 15th)**
Begin comparing the initial results with those from traditional LLMs and other state-of-the-art diagnostic models.
- **Fine-tuning of HyperBERT model (Timeline: February 16th – March 1st)**
Optimize the model to produce the most accurate results possible, while ensuring that it provides concepts to support its findings.
- **Finalizing Results: Comparison between basic LLM and RAG HyperBERT for the same symptoms (Timeline: March 2nd – March 15th)**
Evaluate the fine-tuned model’s performance against state-of-the-art models to highlight potential benefits, and document reportable results.
- **Interpretation Analysis (Timeline: March 16th – April 12th)**
Analyze how the model interpreted symptoms and examine the relationships it recognized in detail to determine if previously unknown biological functions are revealed.
- **Sensitivity Analysis between benchmark model and RAG HyperBERT model (Timeline: April 12th – April 26th)**
Conduct sensitivity tests to assess model robustness and performance under varying conditions, including increasingly vague diagnosis.
- **Contingencies (Timeline: April 27th – May 3rd)**
Outline alternative strategies to address potential challenges or adjustments to the initial plan.
- **Dissertation Writing (Timeline: May 3rd – hand-in date)**
Begin the final phase of dissertation writing.

3 References

References

- [1] Adrián Bazaga, Pietro Liò, and Gos Micklem. Hyperbert: Mixing hypergraph-aware layers with language models for node classification on text-attributed hypergraphs, 2024.
- [2] Ayer A. Ergün C, Urhan M. A review on the relationship between gluten and schizophrenia: Is gluten the cause?, 2018 Sep;21(7):455-466.
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- [5] Han Xuanyuan, Pietro Barbiero, Dobrik Georgiev, Lucie Charlotte Magister, and Pietro Lió. Global concept-based interpretability for graph neural networks via neuron analysis, 2023.